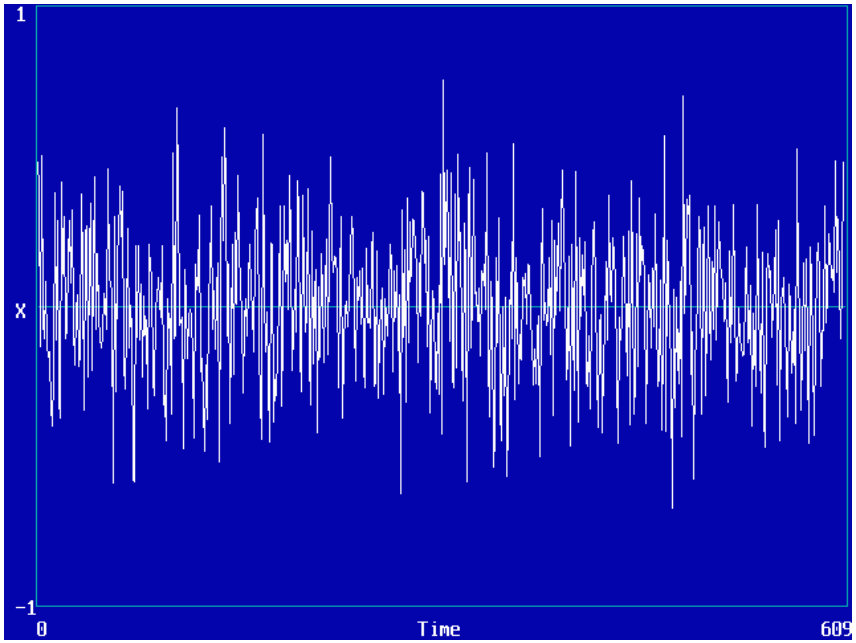
Instrumentation Filters

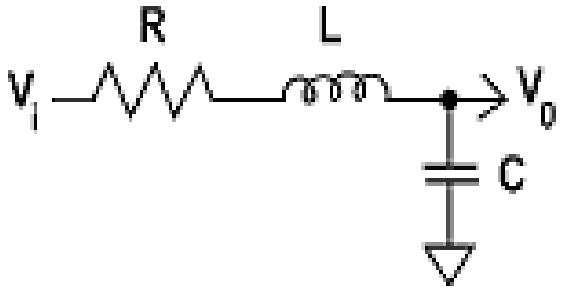
Analogue Filters

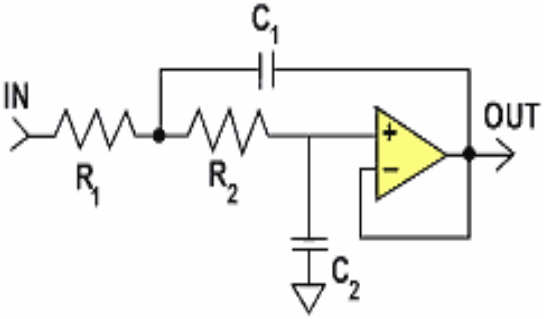
- Described by their pass band
- Filter's pass band is the range of frequencies over which it will pass an incoming signal
- Signal frequencies lying outside the pass band are attenuated



Types of Filters

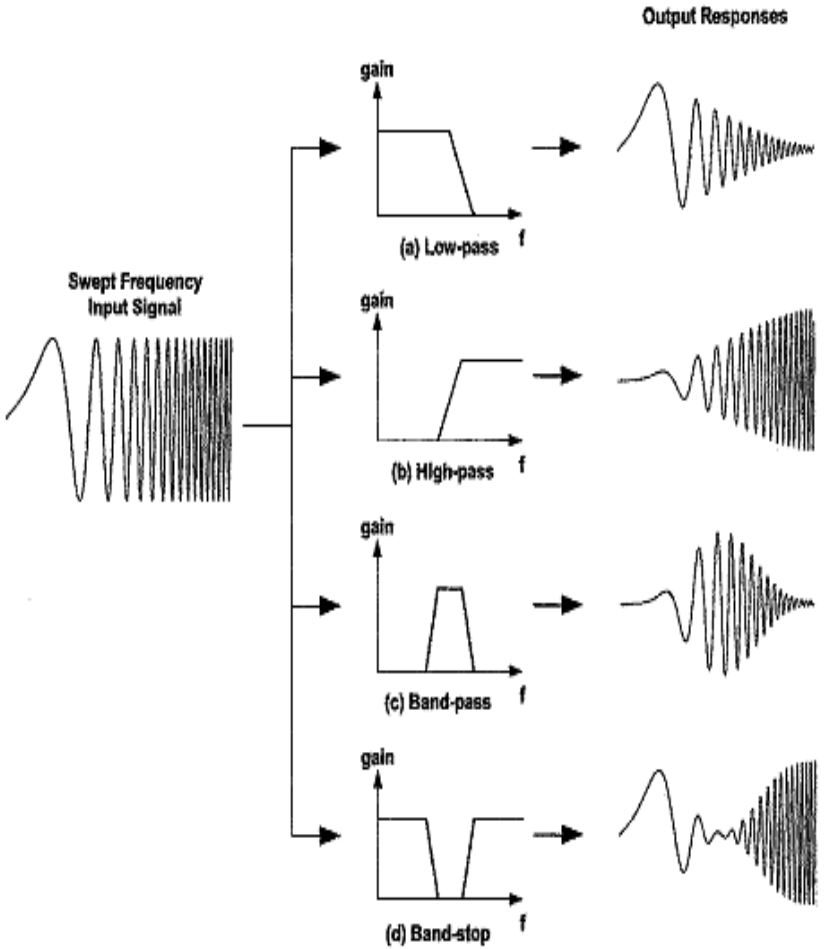
- Passive Filters
  - Do not contain amplifying devices to strengthen the signal
  - the output signal has a smaller amplitude than its corresponding input signal
  - Does not need power
- Active Filters
  - contain amplifying devices to increase signal strength
  - Need to be powered to work





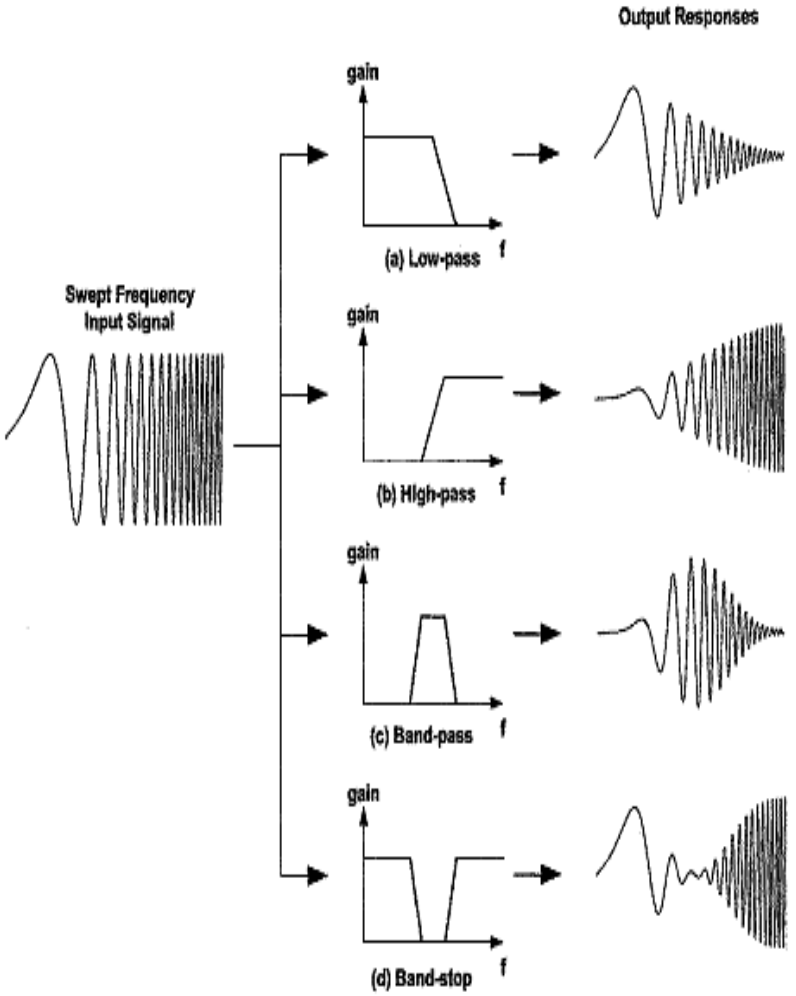
Filters classification

- *Low-pass filters*
  - pass low-frequency signals while blocking high-frequency signals
  - The pass band ranges from DC (0 Hz) to a corner frequency F_C
- *High-pass filters*
  - pass high-frequency signals while blocking DC and low-frequency signals
  - The pass band ranges from a corner frequency (F_C) to infinity



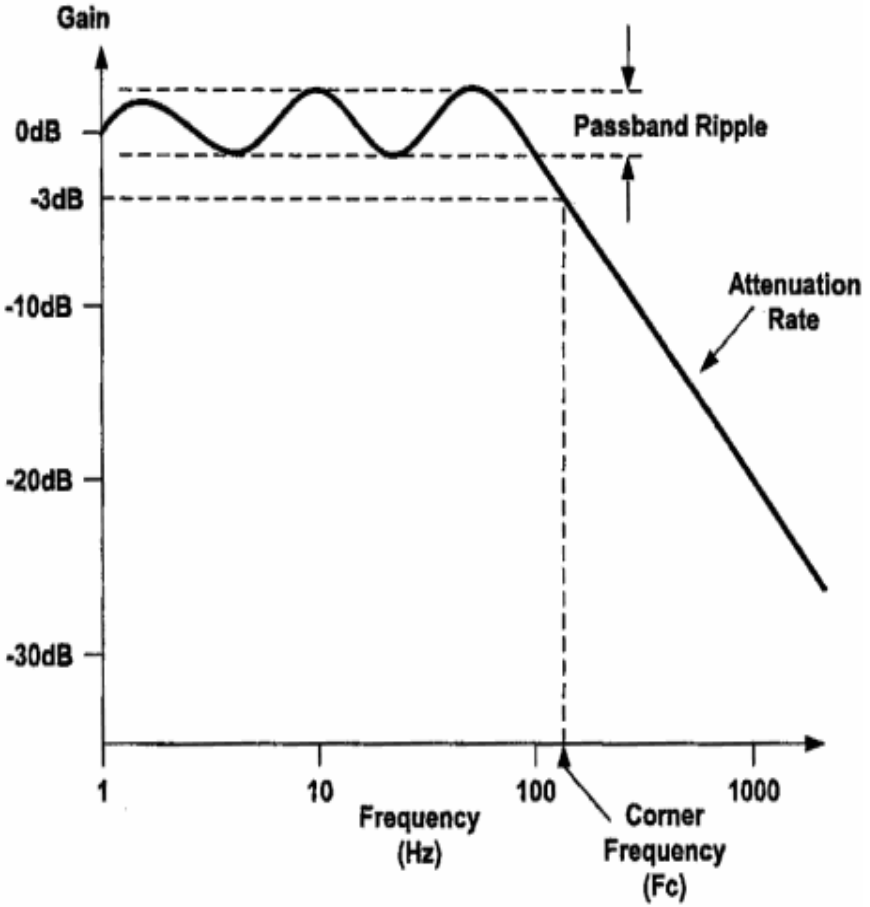
Filters classification(Cont…)

- *Band-pass* filter
  - pass only signals between two given frequencies, blocking lower and higher
  - The pass band lies between two frequencies, F_L and F_H. Signals between DC and F_L are blocked, as are signals from F_H to infinity
- Band-stop filters
  - Also known a notch filters
  - block signals occurring between two given frequencies, F_L and F_H



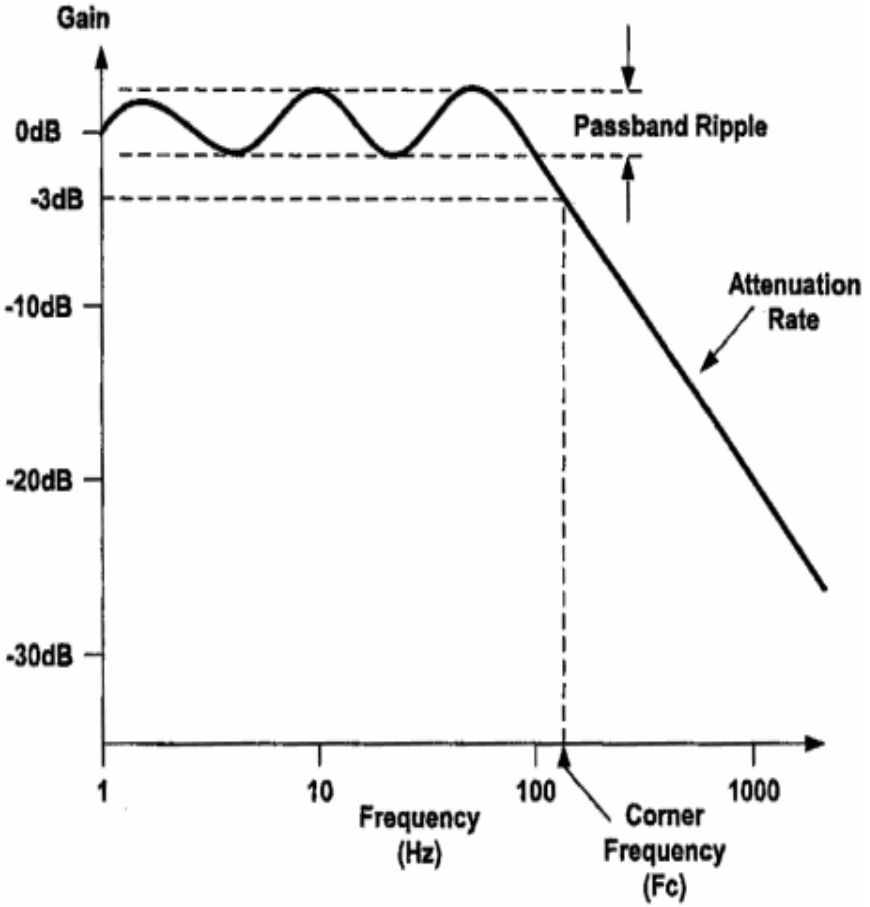
Bode Plots

- Mostly used to plot filter response
- describe the behavior of a filter by relating the magnitude of the filter's response (gain) to its frequency
- The decibels of gain of a filter relate to the ratio between input and output voltages:
- The horizontal axis represents frequency, measured in hertz (Hz), and the vertical axis is measured in decibels (dB).



Bode Plots (Cont…)

- Corner (Cut-off) Frequency
  - the frequency at which the response is 1/2 (0.707) of that in the pass band
  - also referred to as the –3 dB point.
- Attenuation Rate
  - The transition between the pass band and the stop band is a continuous function, and the rate at which this transition occurs
- Pass-Band Ripple
  - The magnitude of the response may vary inside the pass band



Design of Passive Filters

- The designed are based on the fact that:
  - Resistor has a constant impedance over frequency,
  - The impedance of a capacitor decreases with increasing frequency
  - The impedance of an inductor increases with increasing frequency

$$Z_R = R$$

$$X_C = \frac{1}{2\pi f C}$$

$$X_L = 2\pi f L$$

Low pass Filter (1st order)

- In series connection the output voltage is given by

$$V_{out} = V_{IN}\left(\frac{R_1}{R_1 + R_2}\right)$$

- But total impedance for resistor and capacitor in series is given as

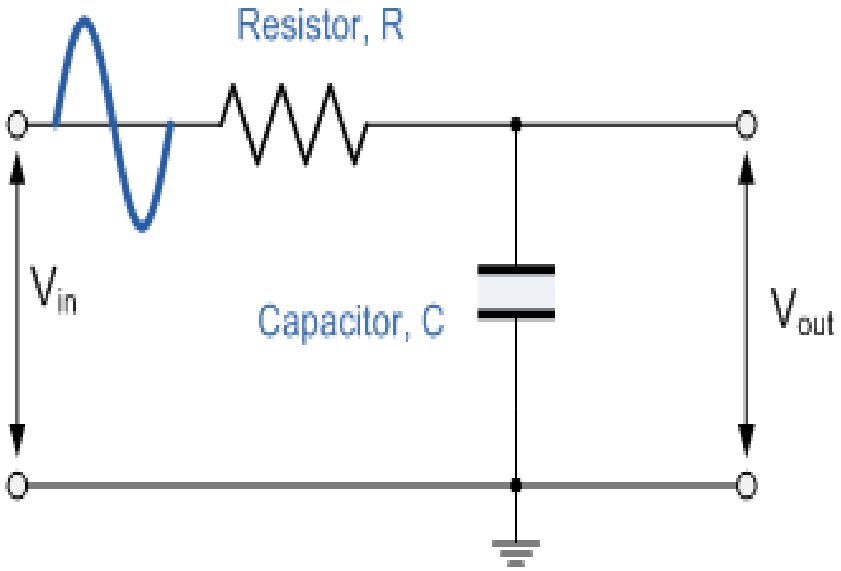
$$Z_T = \sqrt{R^2 + {X_C}^2}$$

$$\frac{V_{out}}{V_{IN}} = \frac{X_C}{\sqrt{R^2 + {X_C}^2}}$$

- Cut-off frequency happen when

$$R = X_C \quad \Rightarrow \quad f_c = \frac{1}{2\pi RC}$$

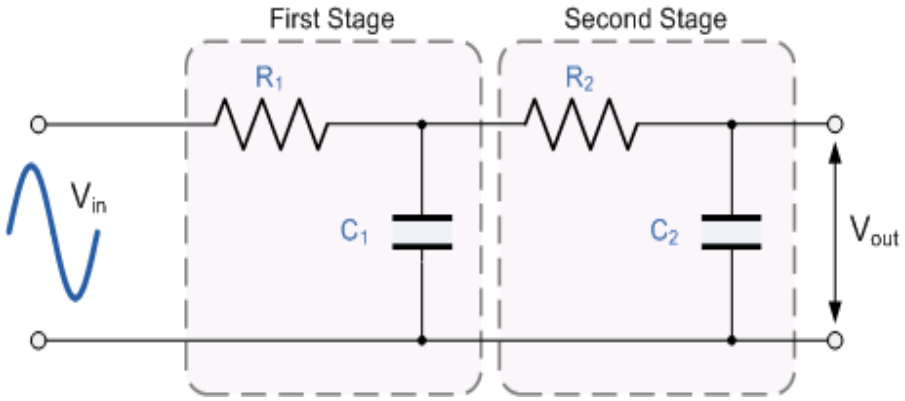
- Low pass Example

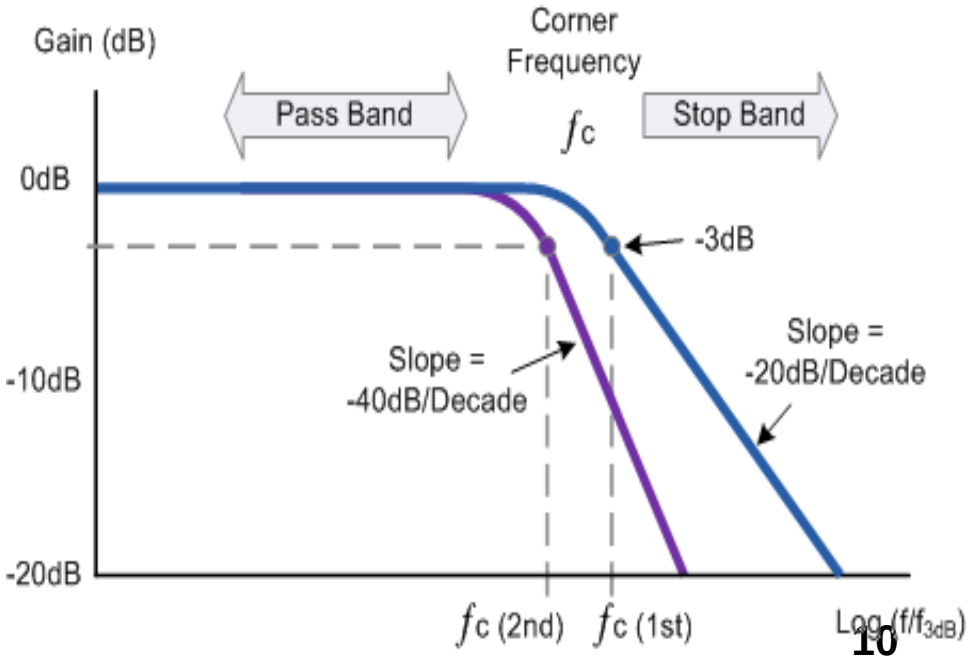


Low pass Filter (2nd order)

- A first-order low pass filter can be converted into a second-order type by simply using an additional RC network
- If a number (*n*) of such filters are cascaded together, the resulting filter circuit would be known as an "*n*th-order" filter with a slope of *n* x -20dB/decade
- Cut-off frequency then becomes

$$f_C = \frac{1}{2\pi\sqrt{R_1 R_2 C_1 C_2}} Hz$$





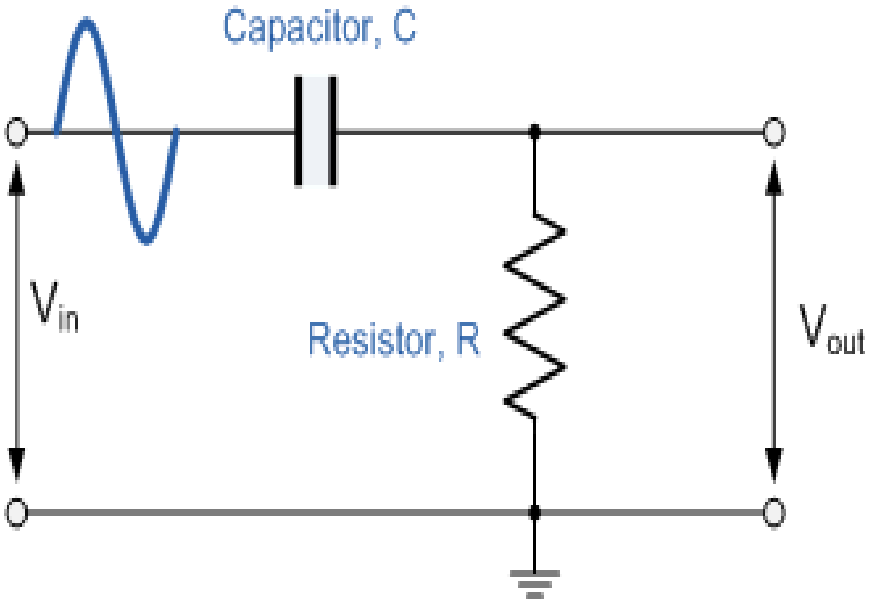
High pass Filter (1st order)

- The exact opposite to the low pass filter

$$\frac{V_{out}}{V_{IN}} = \frac{R}{\sqrt{R^2 + {X_C}^2}}$$

- Only passes signals above the selected cut-off point, fc eliminating any low frequency signals
- Application:
  - audio amplifier in speaker systems
- Cut-off frequency happen when

$$R = X_C \Rightarrow f_c = \frac{1}{2\pi RC}$$

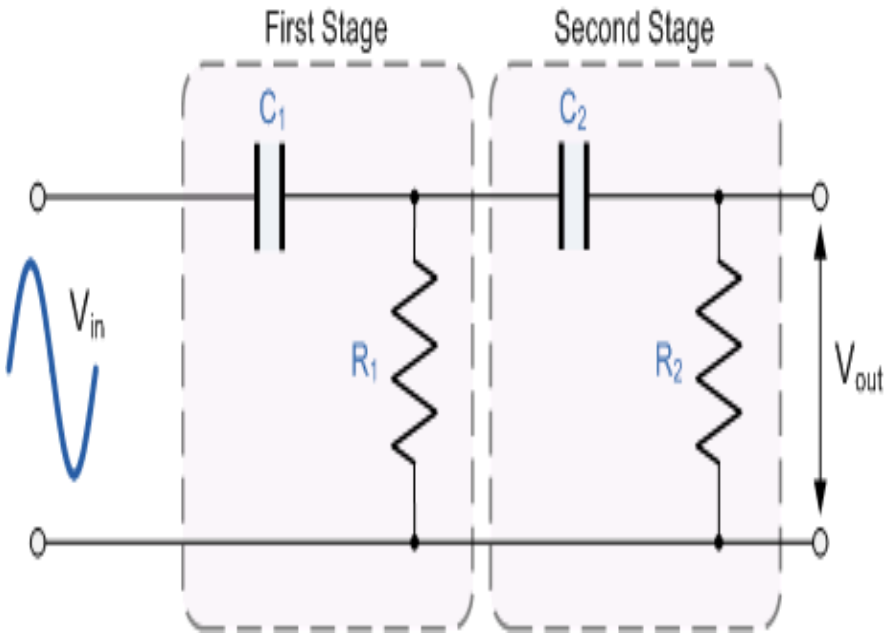


High pass Filter (2nd order)

- Two first-order high pass filters connected or cascaded together to form a second-order or two-pole filter network

- Cut-off frequency then becomes

$$f_C = \frac{1}{2\pi\sqrt{R_1 R_2 C_1 C_2}} Hz$$



Cascaded Band pass Filter

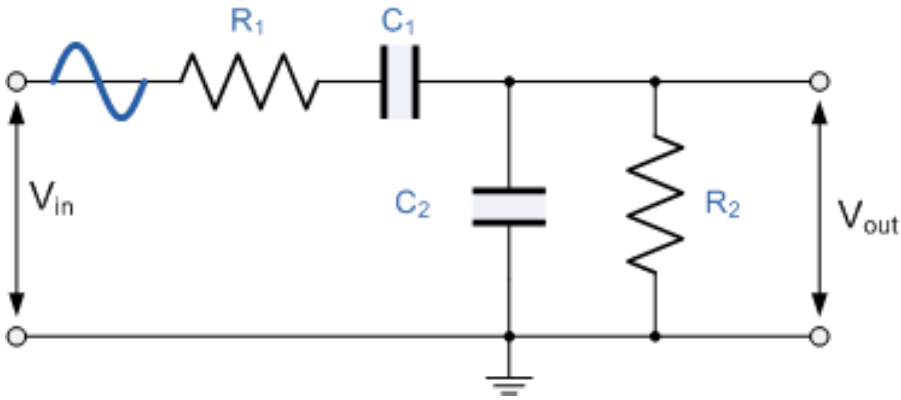
- Formed by cascading together a single low pass filter with a high pass filter
- Has a pass band known as Bandwidth given by

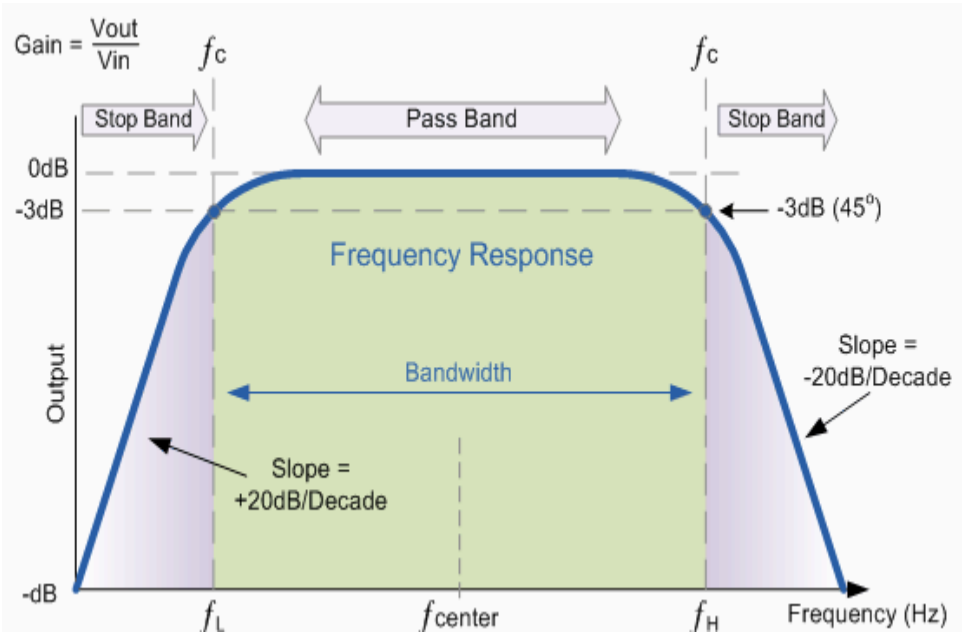
$$BW = f_H - f_L$$

- Resonant or Centre Frequency
  - Frequency where the output gain is at its maximum or peak value

$$f_r = \sqrt{f_H \times f_L}$$

- Band pass Example





RLC Band pass Filter

- Formed by connecting RLC in series
- The cut-off frequencies are at the –3 dB half-power points

$$f_L = -\frac{R}{2L} + \sqrt{(\frac{R}{2L})^2 + \frac{1}{LC}}$$

$$f_H = \frac{R}{2L} + \sqrt{(\frac{R}{2L})^2 + \frac{1}{LC}}$$

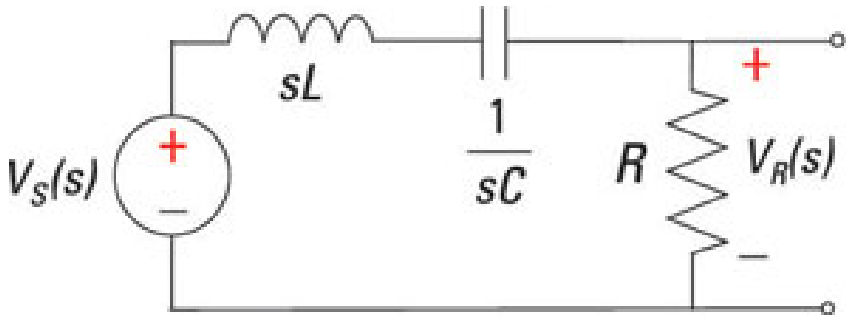
$$BW = f_H - f_L = \frac{R}{L}$$

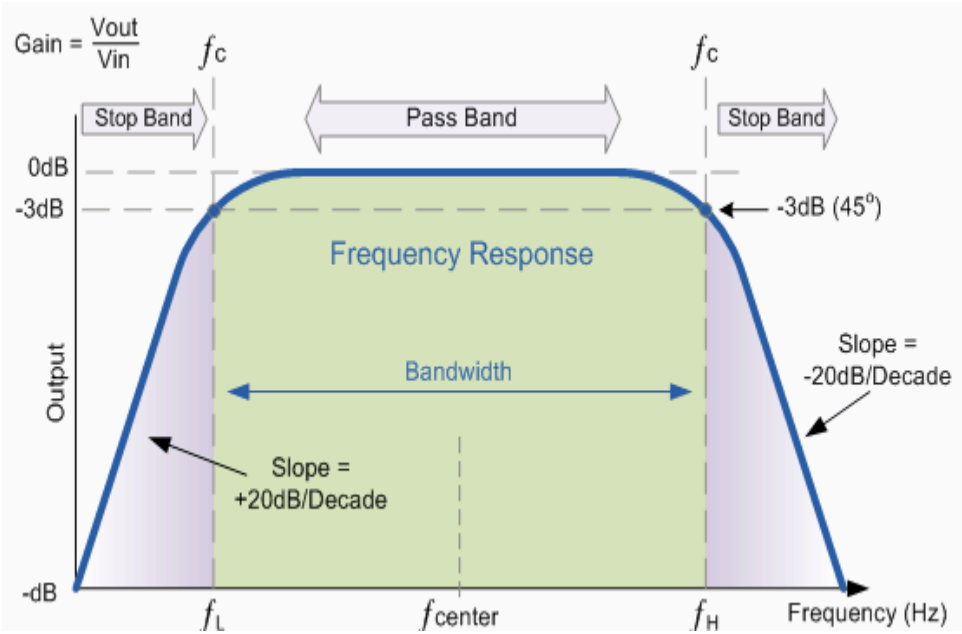
- The Quality factor, Q, is defined as the ratio of center frequency against bandwith

$$Q = \frac{f_r}{BW} = \frac{1}{R}\sqrt{\frac{L}{C}}$$

- The resonant frequency

$$f_r = \sqrt{f_H \times f_L} = \frac{1}{\sqrt{LC}}$$





Design of Active Filters

- Uses similar concepts as in passive design
- Introduces the use of operational amplifiers
- Hence can amplify input signals

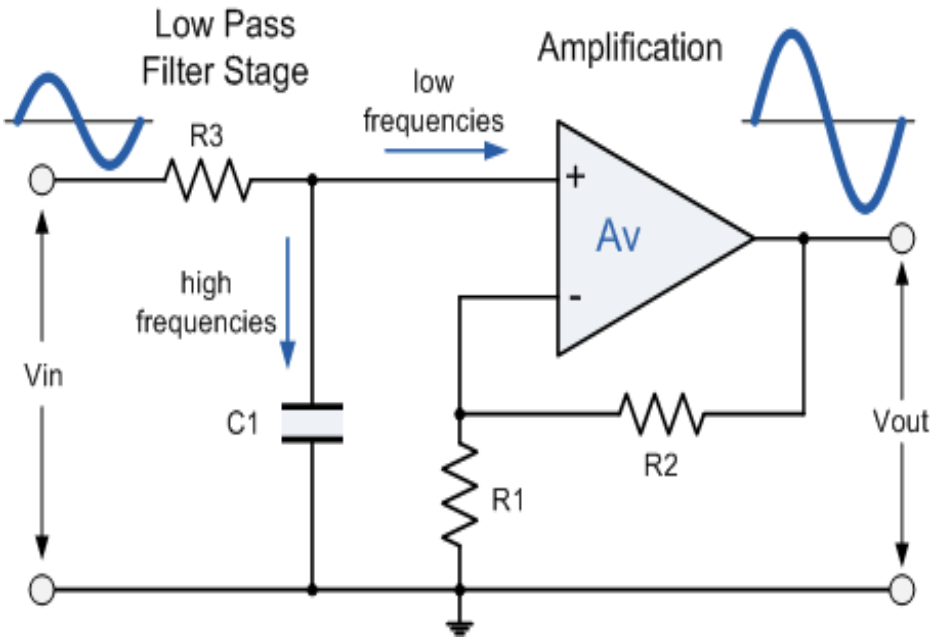
Active Low pass Filter

- Frequency response is exactly the same as low-pass passive filter
- uses an op-amp for amplification and gain control

- For a non-inverting amplifier circuit, the magnitude of the voltage gain for the filter is given as

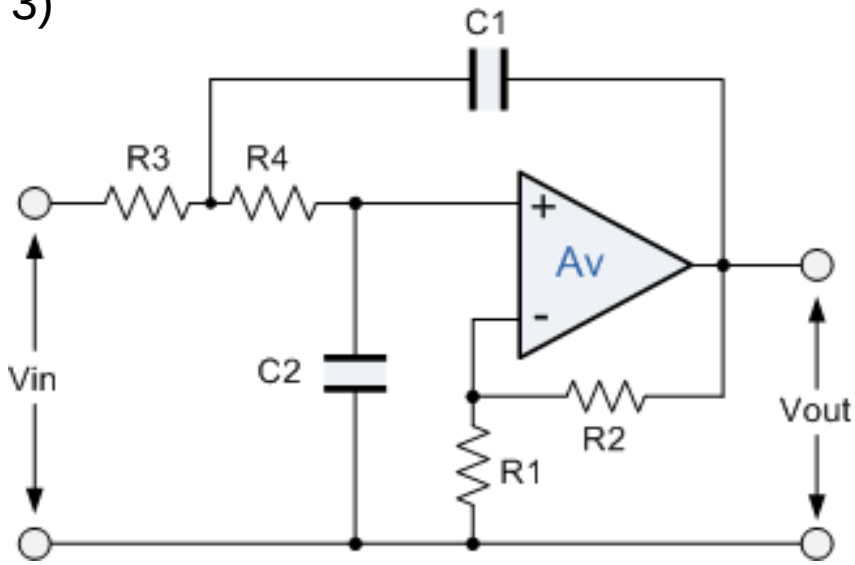
$$dcgain = (1 + \frac{R_2}{R_1})$$

- Active low pass example



Second-order Low Pass Active Filter

- Formed by adding another RC network in the input path
- Similarly roll-off will be twice the first-order filters
- Sometimes called Salley-key Butterworth Filters
- EE3300 Lab (EMI 3)



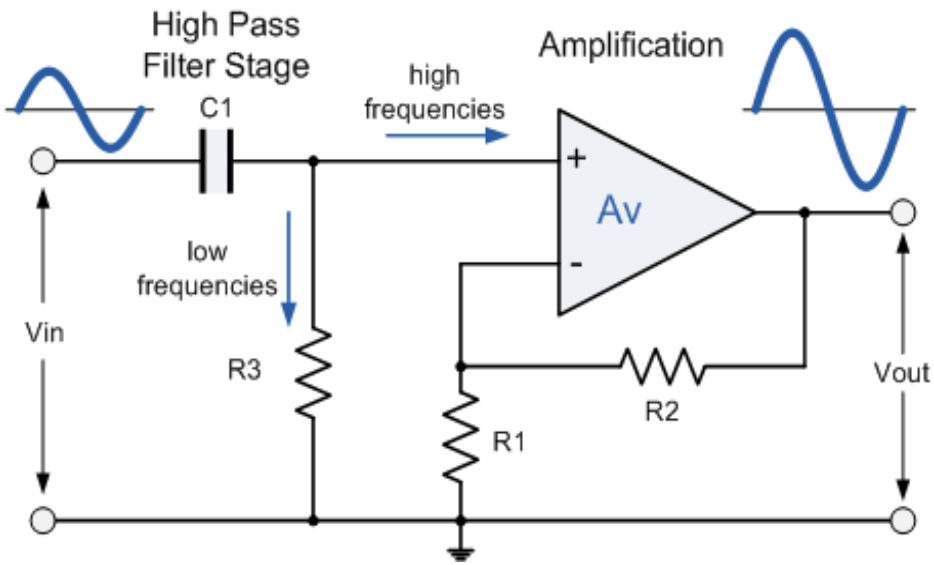
$$Av = 1 + \frac{R_2}{R_1}$$

$$f_C = \frac{1}{2\pi\sqrt{R_3R_4C_1C_2}}$$

Active High pass Filter

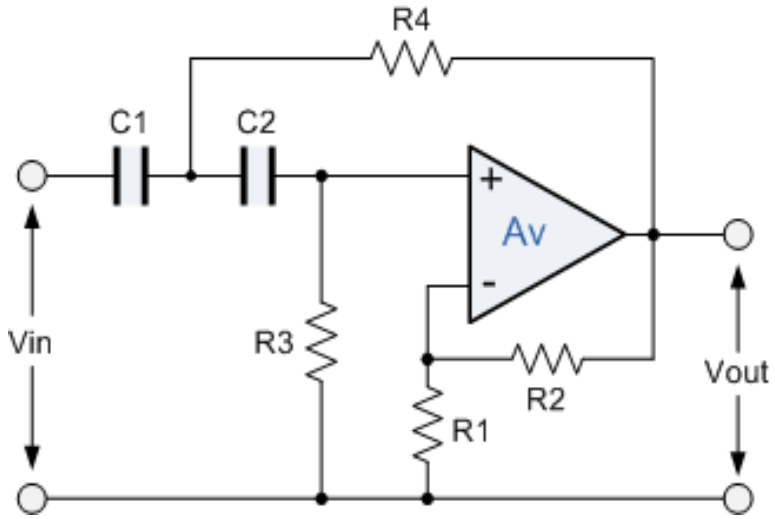
- Frequency response is exactly the same as high-pass passive filter
- Attenuates low frequencies and passes high frequency signals
- For a non-inverting amplifier circuit, the magnitude of the voltage gain for the filter is given as

$$dcgain = (1 + \frac{R_2}{R_1})$$



Second-order High Pass Active Filter

- Formed by adding another RC network in the input path
- Similarly roll-off will be twice the first-order filters
- Also called Salley-key Butterworth Filters
- EE3300 Lab (EMI 3)



$$Av = 1 + \frac{R_2}{R_1}$$

$$f_C = \frac{1}{2\pi\sqrt{R_3R_4C_1C_2}}$$

- Homework
  - Draw a circuit for 3rd order high pass filter

Active Band pass Filter

- Frequency response is exactly the same as band-pass passive filter



$$dcgain = (1 + \frac{R_4}{R_3})$$

